

AUSTIN L. NASH

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EDUCATION

- Purdue University**, West Lafayette, IN *December 2019*
Ph.D. in Mechanical Engineering (GPA: 3.91/4.0)
Dissertation Title: “Hierarchical Combined Plant and Control Design for Thermal Management Systems”
Adviser: Prof. Neera Jain
- Rose-Hulman Institute of Technology**, Terre Haute, IN *May 2015*
M.S. in Mechanical Engineering (GPA: 4.0/4.0)
Thesis Title: “Euler-Lagrange Optimal Control of Indirect Fire Symmetric Projectiles”
Adviser: Prof. Bradley T. Burchett
- Rose-Hulman Institute of Technology**, Terre Haute, IN *May 2013*
B.S. in Mechanical Engineering, *summa cum laude* (GPA: 3.93/4.0)

RESEARCH INTERESTS

Dynamic Modeling and Control of Energy Systems, Combined Plant and Control Design, Energy Management and Storage, Optimal Control Methods, Robust Control Methods, Constrained Optimization

PUBLICATIONS

Journal Articles

- J1. **A. Nash** and N. Jain, “Hierarchical Combined Plant and Control Design: A Case Study on Thermal Management Systems,” *ASME Journal of Mechanical Design*, (submitted).
- J2. **A. Nash** and N. Jain, “Combined Plant and Control Co-design for Robust Disturbance Rejection in Thermal-Fluid Systems,” *IEEE Transactions on Control Systems Technology*, August 2019.
- J3. **A. Nash**, B. Fu, T. Bird, N. Jain, and T. Fisher, “Control-Oriented Modeling of Integrated Flash Boiling for Rapid Transient Heat Dissipation,” *AIAA Journal of Thermophysics and Heat Transfer*, vol. 33, July 2019.
- J4. **A. Nash**, A. Badithela, and N. Jain, “Dynamic Modeling of a Sensible Thermal Energy Storage Tank with an Immersed Coil Heat Exchanger under Three Operation Modes,” *Applied Energy*, vol. 195, pp. 877-889, June 2017.

Conference Proceedings with Full Paper (* indicates invited paper)

- C1. **A. Nash** and N. Jain, “Dynamic Design Optimization for Thermal Management: A Case Study on Shell and Tube Heat Exchangers,” *Proceedings of the 2019 ASME Dynamic Systems and Control Conference*, Park City, UT, October 8-11, 2019.*
- C2. **A. Nash**, T. Fisher, and N. Jain, “Combined Plant and Control Design for a Flash Boiling Cooling System,” *Proceedings of the 2019 American Control Conference*, Philadelphia, PA, July 10-12, 2019.*
- C3. **A. Nash** and N. Jain, “Second Law Modeling and Robust Control for Thermal-Fluid Systems,” *Proceedings of the 2018 ASME Dynamic Systems and Control Conference*, Atlanta, GA, September 30 - October 3, 2018.*
- C4. **A. Nash** and N. Jain, “Dynamic Modeling and Performance Analysis of Sensible Thermal Energy Storage Systems,” *Proceedings of the 4th International High Performance Buildings Conference*, Purdue University, West Lafayette, IN, July 11-14, 2016.
- C5. **A. Nash** and B. Burchett, “Euler-Lagrange Optimal Control of Indirect Fire Symmetric Projectiles,” *Proceedings of the 2016 AIAA Science and Technology Forum and Exposition*, San Diego, CA, January 4-8, 2016.

- C6. B. Burchett and A. Nash, "Euler-Lagrange Optimal Control for Symmetric Projectiles," *Proceedings of the 2015 AIAA Science and Technology Forum and Exposition*, Kissimmee, FL, January 5-9, 2015.

Theses

- T1. A. Nash, "Hierarchical Combined Plant and Control Design for Thermal Management Systems," *Ph.D. Dissertation*, Purdue University, December 2019.
- T2. A. Nash, "Euler-Lagrange Optimal Control of Indirect Fire Symmetric Projectiles," *M.S. Thesis*, Rose-Hulman Institute of Technology, May 2015.

INVITED TALKS

1. A. Nash, "Reimagining Dynamic Energy Management Through Model-Based Hierarchical Control Co-design," *Rose-Hulman Institute of Technology*, Terre Haute, IN, January 6, 2020.
2. A. Nash, "Hierarchical Combined Plant and Control Design for Thermal Management Systems," *Kettering University*, Flint, MI, December 10, 2019.
3. A. Nash, "Combined Plant and Control Design for Thermal-Fluid Systems," *Milwaukee School of Engineering*, Milwaukee, WI, May 10, 2019.
4. A. Nash and N. Jain, "Optimal Plant and Control Co-Design of Thermal Management Systems Subject to Transport Delays," *NATO Science and Technology Organization Workshop: AVT-270/TWS-028 Validation of Thermal Models for Military Power Systems*, Utrecht, Netherlands, October 10, 2017.

RESEARCH EXPERIENCE

Post-doctoral Research Scholar

January 2019 - present

Purdue University

- Responsible for planning and overseeing multiple undergraduate research projects
- Working to experimentally validate HCCD for dynamic thermal management
- Researching integrated aircraft design and control

Graduate Research Assistant, Ph.D. Candidate

August 2015 - December 2019

Purdue University

- Developed and validated control-oriented dynamic model of a sensible thermal energy storage tank with an immersed coil heat exchanger
- Derived a low-order dynamic model a novel two-phase flash boiling cooling apparatus for rapid transient heat dissipation
- Developed a model formulation technique for real-time evaluation and minimization of second law losses for a class of thermal-fluid systems with mass transport delays
- Developed a nested co-design algorithm aimed at intelligent design and control of thermal management systems for improved system efficiency and thermal endurance
- Developed a transient design optimization algorithm for shell and tube heat exchangers which incorporates linearized eigenvalues as a proxy for speed of heat rejection
- Integrated system level control co-design algorithm with component level design optimization to create hierarchical control co-design (HCCD) framework for optimal dynamic energy management
- Validated HCCD framework with high-fidelity thermal modeling software

Graduate Assistant, M.S. Student

August 2013 - May 2015

Rose-Hulman Institute of Technology

- Developed nonlinear and linear dynamic models of a six-degree-of-freedom rigid symmetric projectile with movable canard actuators
- Derived and implemented linear piecewise time-varying optimal controller to guide projectile firings to a downrange target for indirect fire launches
- Validated controller design on nonlinear plant model using a Monte Carlo set of launch angle pairs

Summer Research Intern*June - August 2011, 2012*

Rose-Hulman Ventures

- Modeled and studied thermal dynamics in computing data centers to minimize cooling costs
- Designed thermal measurement system for quantifying data center performance

TEACHING AND MENTORING EXPERIENCE

Ward A. Lambert Graduate Teaching Fellow*August 2018 - May 2019*

Purdue University

- Taught a section of ME200 - Thermodynamics I - to 84 students
- Received overall rating of 4.8/5.0 based on 72 student reviews
- Prepared all notes and quizzes for 40 lectures through the semester
- Assisted in preparation of three midterm exams and comprehensive final exam
- Hosted weekend review sessions open to 420 students before each midterm exam
- Recruited multiple students from my section to get involved with UG research in the Jain Research Lab

Graduate Student Mentor*May - August 2016*

Summer Undergraduate Research Fellowship Program, Purdue University

- Advised visiting undergraduate research assistant in validating thermal energy storage model
- Provided technical feedback for all program deliverables

Laboratory Assistant*November 2014 - February 2015*

Mechatronic Systems (ME430), Rose-Hulman Institute of Technology

- Helped with laboratory assignments for 60 undergraduate students
- Helped solve software and hardware issues during laboratory sessions

Laboratory Assistant*November 2013 - February 2014*

Computer Applications II (ME323), Rose-Hulman Institute of Technology

- Helped with laboratory assignments for 30 undergraduate students
- Helped solve coding issues during laboratory sessions and grade lab assignments

Graduate Student Adviser*November 2013 - May 2014*

Capstone Design I-II, Rose-Hulman Institute of Technology

- Assisted faculty member as a graduate student adviser for a team of senior design students working to minimize noise during hovercraft operation
- Attended weekly meetings and provided technical feedback on all project deliverables

AWARDS AND RECOGNITION

ASME Energy Systems Technical Committee Best Paper Award Finalist*September 2018*

2018 Dynamic Systems and Control Conference

- One of three finalists among papers in multiple energy systems sessions

Ward A. Lambert Graduate Teaching Fellow*August 2018 - May 2019*

Purdue University

- Awarded to Ph.D. candidates with outstanding potential for a career in academia

Ingersoll-Rand Fellowship*August 2015 - July 2016*

Purdue University

- Awarded to Ph.D. students with outstanding academic record upon entry into the Ph.D. program

PROFESSIONAL ACTIVITIES AND SERVICE

Graduate Student Committee*January 2016 - present*

Herrick Laboratories, Purdue University

- Act as liaison between graduate students and Industrial Advisory Committee
- Coordinate and host seminar series with Industrial Advisory Committee members
- Organize student-led events throughout fall and spring semesters

Manuscript Reviewer

- IEEE Conference on Control Technology and Applications, ASME Dynamic Systems and Control Conference, Purdue High Performance Buildings Conference, AIAA Journal of Thermophysics and Heat Transfer

Memberships (* *indicates student member*)

- American Institute of Aeronautics and Astronautics*
- American Society of Heating, Refrigerating and Air-Conditioning Engineers*
- American Society of Mechanical Engineers*
- Institute of Electrical and Electronics Engineers*